

## STATEMENT OF WORK

### 1. SCOPE

- 1.1. Title: C135 Diesel Heat Exchanger ET Standards.
- 1.2. Location of Work: All to be accomplished at the awarded Vendors location, once they receive the Government Furnished Property.
- 1.3. POP: Shall start immediately following the award & the receipt of the Government supplied material by the vendor, and to be agreed upon by the Government and contractor.

### 2. REFERENCES

- 2.1. NAVSEA Technical Publication T9074-AT-GIB-010/2032 Revision 2, Eddy Current Inspection of Heat Exchanger Tubing on Ships of the United States Navy, Dated 31 January 2012, is the document that controls everything to do with the Eddy Current Inspection of Heat Exchanger Tubing at Portsmouth Naval Shipyard. Our local instructions are written to the requirements of this document. Appendix B, Paragraph B.2c of the document, states, "Reference Standards. Calibration standards shall be fabricated in accordance with Para 4.1.6 and shall be obtained from ZETEC or another NAVSEA-approved source."

### 3. REQUIREMENTS

- 3.1. Vendor shall provide prefabricated defects placed in three separate standards consisting of the same material, in this case "70/30 Copper Nickel alloy (CuNi)", that will be Non-Destructive Testing (NDT) inspected using the Eddy Current method of NDT Inspection.
- 3.2. **Standard #1 - Handheld ASME Type Standard**, Part No.: ASME-022, Description ZCSAA06UOE00.625X0.049-E01-0Z012, Material: 14" Long, 0.625 x 0.049" 70/30 CuNi tubing – Government supplied material (Min. Length required 16").  
**Scope of Flaws (defects) to be placed:**  
**100% TWH, 80%, 60% FBH, 40% FBH, 4 x 20% FBH. Additional 4 x 20% flaw with tube support ring over the flaw.**  
**Tube Support Rings Material: BRASS QQ-B-613 x .75 Wide x 1.25 OD - Qty: 2**  
Government provided material.
- 3.3. **Standard #2 - Handheld EDM Mill Flat Standard**, Part No.: WTMF-027, Description ZCSAD04UOE00.625X0.049-E01-1C012, Material: 36" long, 0.625 x .049" 70/30 CuNi tubing - Government supplied material (Min. Length required 38").  
**Scope of Flaws (defects) to be placed:**  
**Wall Thinning milled flat grooves –**  
**OD @ 10%, 20%, 30%, 40%, 50% & 60%. All Flaws 2 inches long.**  
**ID Radial Wear flaws 20%, 40% & 60% x .25 wide circumferentially x 2 inches long Axially.**  
**Additional 30% OD flaw with ring ½ over flaw.**  
**Tube Support Rings Material: BRASS QQ-B-613 x .75 Wide - Qty: 2** Government provided material.

- 3.4. **Standard #3 - Handheld EDM 360-Degree Ring Standard w/Dual Support Rings**, Part No.: WT360-031, ZCSAD03UOE00.625X0.049-E01-1C032, Material: 34" long, 0.625 x .049" 70/30 CuNi tubing - Government supplied material (Min. Length required 36").

**Scope of Flaws (defects) to be placed:**

**360° OD Wall Thinning grooves - 10%, 20%, 30%, 40% & 50%.**

**360° ID Wall Thinning grooves - 20%, 40% & 60%. All Flaws 2 inches long.**

**Additional 20% OD flaw with Ring ½ over flaw.**

**Tube Support Rings: BRASS QQ-B-613 x .75 Wide - Qty: 2, Government provided material.**

4. **RESPONSIBILITIES**

- 4.1. Portsmouth Naval Shipyard (C135 specifically) shall provide the required Government supplied materials to the vendor:  
30ft of 0.625 x 0.049" 70/30 CuNi tubing.  
Quantity 6 of Tube Support Rings: BRASS QQ-B-613 x 0.75 Wide x 1.25" OD.
- 4.2. Vendor shall provide the service of placing the prefabricated defects into the provided materials, as specified in 3 REQUIREMENTS.
- 4.3. Vendor shall supply NIST traceable Certification Reports stating conformance to applicable specification (s).
- 4.4. Vendor Certification Report shall include As-built dimensions of machined features.

5. **DELIVERY / RETURN**

- 5.1. Government Furnished Property to be shipped from PNSY to Vendor.
- 5.2. Finished Standards to be shipped from Vendor location to PNSY.
- 5.3. Technical Point of Contact:  
**William Seawards**  
**C135T NDT Technical/Training**  
**Building 45 FL2**  
**Portsmouth Naval Shipyard**  
**Kittery, ME 03904**  
**207-438-4690**
- 5.4. Additional Point of Contact:  
**Kevin Masse**  
**C135 Production Controller/Purchaser**  
**Building 45 FL1**  
**Portsmouth Naval Shipyard**  
**Kittery, ME 03904**  
**207-438-1410**